Supplemental Material for "Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes"

Asaf A. Solonnikov and Nadav Katz

The Racah Institute of Physics, The Hebrew University of Jerusalem, Israel

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This Supplemental Material provides the experimental setup and qubit coherence characterization; a precise definition of the finite echo-root-Lorentzian pulse and its resonant cancellation mechanism; the Bloch-equation coherence limit; pulse-cutoff and adiabaticity analyses; the numerical model and numerical protocol comparisons. The final section gives the three-level high-drive analysis: it treats the AC-Stark displacement, accumulated-phase compensation of the operational spectral center, and the limits of the three-level rotating-wave model.

S1. MEASUREMENT SETUP

Experiments were performed on a QPU comprising 11 uncoupled superconducting qubits hosted in a dilution refrigerator. Control and readout were implemented with a Quantum Machines OPX1000 equipped with a microwave front-end module (MW-FEM), together with external cryogenic attenuators, filters, circulators, and amplifiers.

A schematic of the setup is shown in Fig. S1. The MW-FEM uses direct digital synthesis and digital upconversion: a programmed complex pulse envelope is translated to the configured microwave carrier and emitted directly from an RF output, without an external analog local oscillator or IQ mixer [1]. For acquisition, the returned microwave signal is digitally downconverted to a complex baseband stream. QUA demodulation then applies cosine and sine integration weights to the sampled signal and accumulates the result to obtain the measured quadratures [2]. Thus, separate MW-FEM outputs provide the qubit-drive and resonator-readout tones, while an MW-FEM input acquires the returning readout signal. A dedicated DC bias line was available but was not used in these measurements; consequently, the qubits were not necessarily operated at their flux sweet spots.

The device was mounted at the mixing-chamber stage (∼9 mK), with the drive and readout lines attenuated across the refrigerator stages. The returning readout signal passed through circulators and a 4 K high-electron-mobility transistor (HEMT) amplifier before entering the MW-FEM for digital downconversion and demodulation.

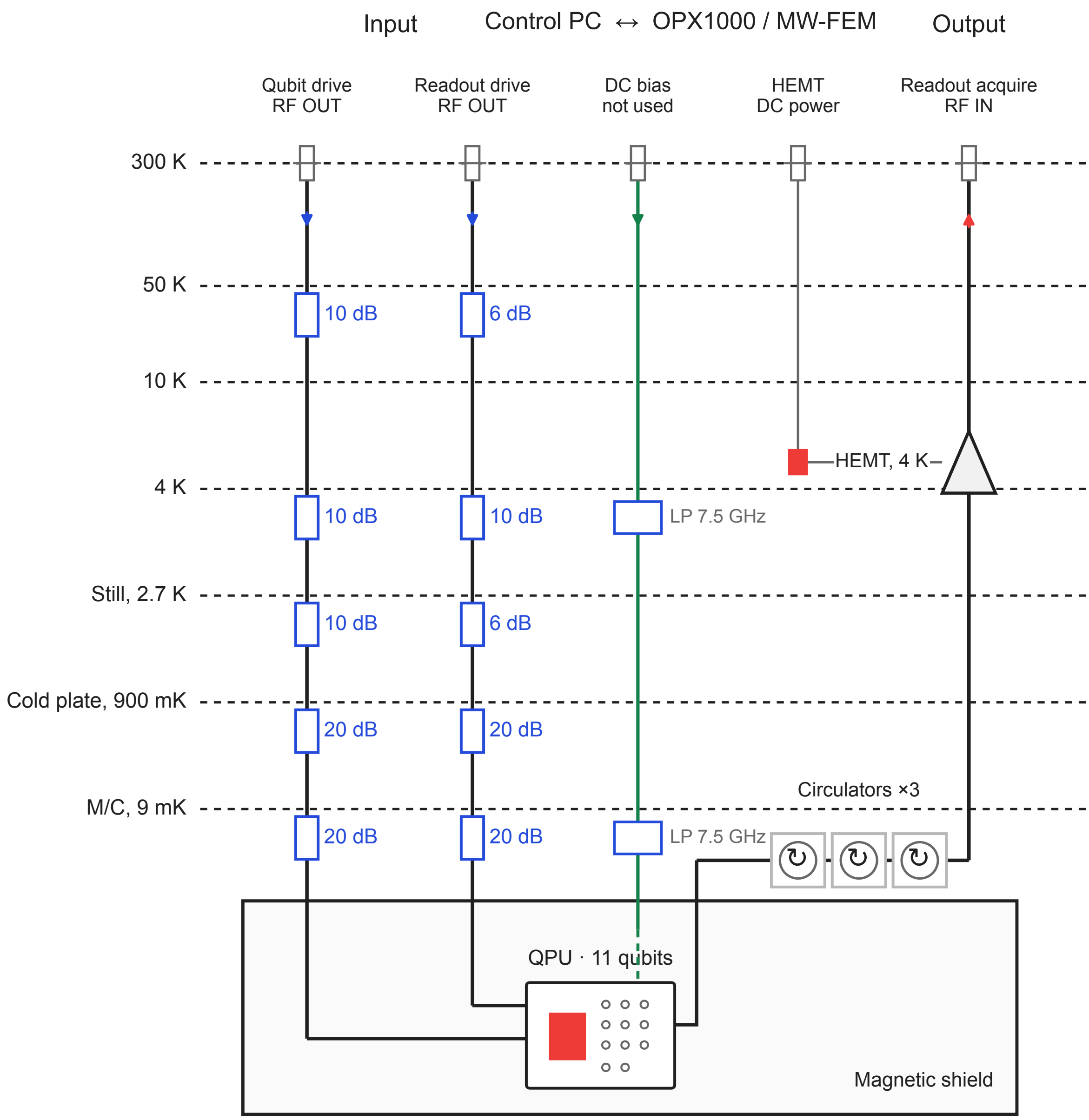

Figure S1. Schematic of the experimental setup for qubit control and readout. A control computer submits QUA programs and receives data over Ethernet. The OPX1000 MW-FEM directly generates the qubit-drive and resonator-readout RF signals and acquires the returning readout signal. Input lines are attenuated across the 300 K, 50 K, 10 K, 4 K, still, cold-plate, and mixing-chamber stages. The low-pass-filtered DC-bias line is shown for completeness but was not used. At the mixing chamber, the device is isolated by circulators; the returning readout signal is amplified by a 4 K HEMT before entering the MW-FEM for digital downconversion and demodulation.

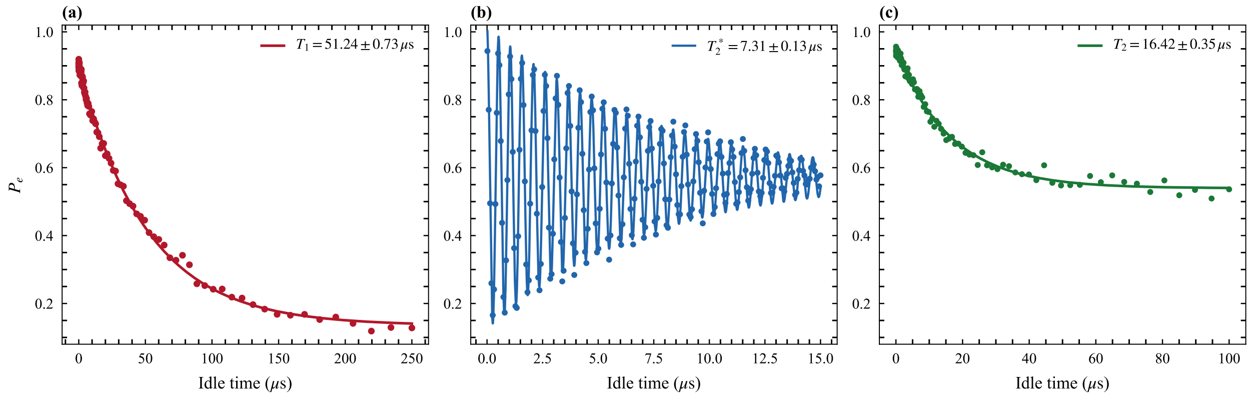

Figure S2. Independent coherence characterization of qubit q1. (a) Energy relaxation following state preparation, together with an exponential fit. (b) Ramsey fringes acquired with a positive 2 MHz programmed detuning, together with an exponentially damped sinusoidal fit. (c) Hahn-echo decay with an exponential fit.

S1.1. Qubit coherence characterization

The energy-relaxation and Ramsey-dephasing times of the qubit used for the spectroscopy measurements were characterized independently. Fig. S2 shows representative measurements for qubit q1. Fitting the excited-state population to an exponential decay with a constant offset gives $T_1 = 51.2\ \mu$s. Fitting the positive-detuning Ramsey trace to an exponentially damped sinusoid gives $T_2^* = 7.3\ \mu$s. A Hahn-echo measurement gives $T_2 = 16.4\ \mu$s from an exponential fit with a constant offset. The quoted uncertainties are one-standard-deviation estimates from the fit covariance matrices.

S2. FINITE PULSE DEFINITION AND ECHO CANCELLATION

We define the finite generalized Lorentzian envelope by

$$f_{n,c,L}(t) = \begin{cases} \left[1 + (t/\tau)^2\right]^{-n}, & |t| \le L/2, \\ 0, & |t| > L/2, \end{cases} \tag{S1}$$

where L is the total pulse duration, n is the shape exponent, and the relative endpoint amplitude is $c = f_{n,c,L}(L/2)$. These parameters imply

$$\tau = \frac{L/2}{\sqrt{c^{-1/n} - 1}}. \tag{S2}$$

The root-Lorentzian pulse used here corresponds to $n = 1/2$. Its echo version has a phase inversion at the center,

$$\Omega_{\mathrm{E}}(t) = \Omega_0 f_{n,c,L}(t) s(t), \qquad s(t) = \begin{cases} +1, & t < 0, \\ -1, & t \geq 0. \end{cases} \tag{S3}$$

The value assigned at the single point $t = 0$ has no effect in the continuum limit. In a sampled waveform, the two halves are constructed with equal duration and opposite phase.

Two cutoff scans must be distinguished. A pure truncation scan keeps τ, n, and Ω_0 fixed while changing L. A fixed-duration hardware scan instead keeps L fixed and changes the endpoint ratio c; in the parametrization of Eq. (S2), this also changes τ. Results from these two scans need not coincide.

In the rotating-wave approximation, the unitary two-level Hamiltonian is

$$\frac{H(t)}{\hbar} = \frac{1}{2} \left[\Delta \sigma_z + \Omega_{\mathrm{E}}(t) \sigma_x \right]. \tag{S4}$$

At exact resonance, Hamiltonians at different times are proportional to the same operator and therefore commute. The propagator is

$$U(\Delta = 0) = \exp\left[-\frac{i\sigma_x}{2} \int_{-L/2}^{L/2} \Omega_{\mathrm{E}}(t)\, dt \right] = \mathbb{I}, \tag{S5}$$

because the signed areas of the two symmetric halves cancel. For $\Delta \neq 0$, the σ_z and σ_x terms do not commute, so the cancellation is incomplete and a narrow detuning-dependent response remains. Equation (S5) is exact only for a symmetric, unitary two-level pulse. Relaxation, dephasing, phase or amplitude imbalance, and finite waveform bandwidth prevent exact cancellation in the experiment.

S3. BLOCH-EQUATION COHERENCE LIMIT

For comparison with the shaped-pulse protocol, we first summarize the steady-state response of a continuously driven two-level system. We define $w \equiv \rho_{00} - \rho_{11}$, with $|0\rangle$ the ground state and $|1\rangle$ the excited state. Hence $P_e = \rho_{11} = (1 - w)/2$; no Pauli matrix sign convention is left implicit. Solving the optical Bloch equations for a drive with angular Rabi frequency Ω_0, angular detuning Δ, relaxation time T_1, and transverse coherence time T_2

gives [3]

$$w_{\rm ss}(\Delta) = \frac{1+\Delta^2 T_2^2}{1+\Delta^2 T_2^2+\Omega_0^2 T_1 T_2}, \qquad P_{e,\rm ss}(\Delta) = \frac{1}{2}\frac{s}{1+s+\Delta^2 T_2^2}, \quad s \equiv \Omega_0^2 T_1 T_2. \tag{S6}$$

This expression assumes a pulse long enough to approach steady state. It is a reference result for conventional saturation spectroscopy, rather than a closed-form description of the finite echo pulse.

At resonance, $P_{e,\rm ss}(0) = s/[2(1+s)]$, which approaches the saturated value $1/2$ only for $s \gg 1$. The positive half-maximum detuning is the angular-frequency HWHM

$$\Delta_{\rm HWHM} = \frac{1}{T_2}\sqrt{1+s}. \tag{S7}$$

The angular-frequency FWHM is therefore

$$\Gamma_\omega \equiv 2\Delta_{\rm HWHM} = \frac{2}{T_2}\sqrt{1+s}. \tag{S8}$$

The corresponding cyclic-frequency FWHM plotted in the Letter is

$$\Gamma_f \equiv \frac{\Gamma_\omega}{2\pi} = \frac{1}{\pi T_2}\sqrt{1+s}. \tag{S9}$$

Thus $\Delta_{\rm HWHM}$, Γ_ω, and Γ_f are distinct quantities and are not denoted by a common symbol.

In the weak-drive limit $s \ll 1$,

$$\Gamma_{\omega,T_2} = \frac{2}{T_2}, \qquad \Gamma_{f,T_2} = \frac{1}{\pi T_2}, \tag{S10}$$

which are the same coherence-limited FWHM expressed in angular frequency and hertz, respectively. In the strong-drive limit $s \gg 1$,

$$\Gamma_\omega \longrightarrow 2\Omega_0\sqrt{\frac{T_1}{T_2}}, \qquad \Gamma_f \longrightarrow 2f_R\sqrt{\frac{T_1}{T_2}}, \tag{S11}$$

where $f_R = \Omega_0/(2\pi)$. These limits are governed by s, not by comparing Ω_0 with an ambiguously defined linewidth.

S4. PULSE TRUNCATION AND ENDPOINT AMPLITUDE

The ideal Lorentzian tail has infinite support, whereas the implemented pulse starts and ends at finite amplitude $\Omega_c = c\Omega_0$. Abruptly switching this residual drive produces an endpoint transient. A useful estimate of the associated population contribution is [4]

$$P_c(\Delta) \simeq \frac{\Omega_c^2}{\Omega_c^2+\Delta^2}\left[1-P_0(\Delta)\right], \qquad \Omega_c = c\Omega_0, \tag{S12}$$

where P_0 denotes the response that would be obtained without the endpoint jump. The artifact becomes important when Ω_c is comparable to the linewidth scale of interest. Thus the cutoff cannot be assessed from c alone: it must be considered together with the peak drive and the desired resolution.

The Bloch-equation reference gives the dimensionless endpoint saturation parameter

$$s_c = \Omega_c^2 T_1 T_2. \tag{S13}$$

Keeping the endpoint contribution in the weak-saturation regime requires

$$s_c \ll 1 \quad \Longleftrightarrow \quad \Omega_c \ll \frac{1}{\sqrt{T_1 T_2}} \quad \Longleftrightarrow \quad c \ll \frac{1}{\Omega_0 \sqrt{T_1 T_2}}. \tag{S14}$$

The inequality therefore bounds the endpoint amplitude from above. It is a perturbative design criterion, not a claim that the cutoff artifact has the steady-state Lorentzian lineshape.

We therefore measured the cutoff dependence over two complementary q1 echo–root–Lorentzian scans. The broad survey spans $c = 0.002$–0.99 with 25 peak-amplitude settings, while the focused sweep resolves $c = 0.01$–0.10 with 200 peak-amplitude settings. Both use a 20 μs pulse and the same measured transverse-coherence reference, $T_2 = 6.856$ μs. We characterize the scans using the dimensionless reciprocal-linewidth metric

$$\mathcal{R} = \frac{\Gamma_{f,T_2}}{\Gamma_f} = \frac{1/(\pi T_2)}{\Gamma_f}, \tag{S15}$$

where every linewidth is a cyclic-frequency FWHM in hertz. Thus $\mathcal{R} = 1$ is the T_2-limited linewidth and $\mathcal{R} < 1$ denotes a broader feature. We also use the signal-weighted metric $|A_{\text{fit}}|\mathcal{R}$, which suppresses narrow fits with negligible contrast. These are linewidth-based scores, not direct estimates of frequency-estimation uncertainty.

The analysis retains only resolved and bounded probability features: the fitted contrast is between 0.05 and 1, $R^2 \geq 0.60$, the fitted center lies inside the measured frequency interval, and the FWHM lies between two frequency bins and half the sweep span. Measurements that do not pass this common screen are omitted. The denser focused sweep provides substantially more accepted fits than the coarse survey and resolves the loss of reciprocal-linewidth performance as the cutoff increases across the 0.01–0.10 interval.

S5. ADIABATIC-BASIS DESCRIPTION

For one half of the pulse, the rotating-frame Hamiltonian can be written

$$\frac{H(t)}{\hbar} = \frac{1}{2}\begin{bmatrix} \Delta & \Omega(t) \\ \Omega(t) & -\Delta \end{bmatrix}. \tag{S16}$$

Define

$$\epsilon(t) = \sqrt{\Delta^2 + \Omega^2(t)}, \qquad \vartheta(t) = \frac{1}{2}\,\mathrm{atan2}[\Omega(t), \Delta]. \tag{S17}$$

Both ϵ and $\dot{\vartheta}$ have units of inverse seconds, whereas ϑ is dimensionless. The instantaneous eigenvalues are $E_- = -\hbar\epsilon/2$ and $E_+ = +\hbar\epsilon/2$. Taking the instantaneous eigenvectors as the columns of

$$R(t) = \begin{bmatrix} -\sin\vartheta & \cos\vartheta \\ \cos\vartheta & \sin\vartheta \end{bmatrix}, \tag{S18}$$

the transformed Hamiltonian is $H_a = R^\mathsf{T} H R - i\hbar R^\mathsf{T}\dot{R}$. Therefore

$$\frac{H_a(t)}{\hbar} = \begin{bmatrix} -\epsilon/2 & -i\dot{\vartheta} \\ i\dot{\vartheta} & \epsilon/2 \end{bmatrix}, \qquad \dot{\vartheta} = \frac{\Delta\dot{\Omega}}{2(\Delta^2 + \Omega^2)}. \tag{S19}$$

The nonadiabatic coupling is $|\dot{\vartheta}|$, while the instantaneous angular-frequency gap is ϵ. Adiabatic following therefore requires [5]

$$|\dot{\vartheta}| \ll \epsilon. \tag{S20}$$

Replacing the inequality by equality provides the local crossover estimate

$$\sqrt{\Delta_b^2 + \Omega^2(t_0)} = \frac{|\Delta_b \dot{\Omega}(t_0)|}{2[\Delta_b^2 + \Omega^2(t_0)]}, \tag{S21}$$

where t_0 is the time at which the nonadiabatic coupling is largest for the detuning under consideration. This is a scaling criterion, not an exact linewidth formula.

This smooth-basis construction applies separately to each half of the echo pulse; the imposed phase jump at $t = 0$ is a control discontinuity and is not represented by $\dot{\vartheta}$ in this local adiabatic estimate.

For $L^{(n)}(t) \sim |t|^{-2n}$, the inverse-power-tail derivation of Boradjiev and Vitanov requires $2n > 1$. For an untruncated generalized Lorentzian with $n > 1/2$, it gives

$$\Delta_b \tau \propto (\Omega_0 \tau)^{-1/(2n-1)}. \tag{S22}$$

As $n \to 1/2^+$, the exponent $1/(2n-1)$ diverges, indicating the strongest power-narrowing trend within the asymptotic family. The exact root-Lorentzian case $n = 1/2$ is nevertheless marginal and lies outside the domain of Eq. (S22); substituting $n = 1/2$ into that power law is invalid. In particular, the singularity does not imply zero FWHM. The finite pulse length, nonzero endpoint amplitude, decoherence, and echo phase inversion set the observed Γ_f, which is evaluated from the complete simulated and experimental response in this work.

S6. NUMERICAL MODEL

Unless otherwise noted, the simulations use an effective driven two-level system coupled to Markovian relaxation and dephasing channels. Main-text Fig. 2 instead uses the three-state transmon extension described below. In the frame rotating at the drive frequency, and within the rotating-wave approximation, the Hamiltonian is implemented as [6]

$$\frac{H(t)}{\hbar} = -\Delta a^\dagger a + \frac{\alpha}{2} a^{\dagger 2} a^2 + \frac{\Omega_0}{2} f(t) \left(a + a^\dagger\right), \tag{S23}$$

where $\Delta = \omega_d - \omega_{01}$ is the drive–qubit detuning, Ω_0 is the peak Rabi frequency, α is the anharmonicity, and a is the truncated oscillator lowering operator. In the effective two-level model, $a = |0\rangle\langle 1|$ and the anharmonicity term vanishes. Up to an irrelevant identity term, Eq. (S23) is equivalent to $H/\hbar = (\Delta/2)\sigma_z + (\Omega_0 f/2)\sigma_x$.

The simulations use the finite pulse in Eqs. (S1)–(S3). The overall sign of the echo envelope is a phase convention and does not affect the measured populations.

The density matrix evolves according to the Lindblad master equation

$$\dot{\rho} = -\frac{i}{\hbar}[H(t), \rho] + \mathcal{D}\left[\sqrt{\frac{1}{T_1}}\, a\right]\rho + \mathcal{D}\left[\sqrt{\frac{2}{T_\phi}}\, a^\dagger a\right]\rho, \tag{S24}$$

where $\mathcal{D}[C]\rho = C\rho C^\dagger - \frac{1}{2}\{C^\dagger C, \rho\}$ and

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}. \tag{S25}$$

The q1 calculations use $T_1 = 51.2$ μs and $T_{2,\mathrm{eff}} = 7.31$ μs, corresponding to $T_\phi = 7.87$ μs. The q1 anharmonicity is inferred from repeated OPX1000 two-photon spectroscopy. With $f_{01} = 4.267106667$ GHz, three scans locate the sharp, interior response at $f_{02}/2 = 4.159106667$ GHz, including a 1 MHz-step confirmation scan. We therefore use $\alpha/(2\pi) = 2(f_{02}/2 - f_{01}) = -216.0$ MHz; the finest scan gives a grid-scale uncertainty of approximately 1.0 MHz for the inferred anharmonicity. Each calculation starts in $|0\rangle$, integrates Eq. (S24) over the complete pulse, and records the final state observables. Spectroscopy maps are generated by repeating this evolution over the experimental detuning and drive-amplitude grids, using the pulse duration, cutoff, T_1, and T_2 associated with the corresponding data set. The time-dependent master equation is solved with QuTiP [7] or, for dense matched sweeps, the algebraically equivalent two-level Bloch equations with fourth-order Runge–Kutta integration. The dense main-text Fig. 2 maps use the states $|g\rangle$, $|e\rangle$, and $|f\rangle$, with $\alpha/(2\pi) = -216.0$ MHz, and report $P_e = \rho_{ee}$ while retaining $P_f = \rho_{ff}$ in the archived figure data. Other production calculations use a two-dimensional Hilbert space and a uniform temporal sampling of the complete waveform. The default solver grid contains 1000 points; the matched linewidth sweep uses 2000 steps. The even grids contain equal numbers of negative- and positive-time samples, with the central phase inversion between them.

Figure S3 quantifies the tradeoff between pulse duration, linewidth, and signal contrast at fixed cutoff $c = 0.005$. The continuous curves show the simulated response, while the markers show the experimental data. Panels (a) and (b) separate spectral narrowing from signal loss: longer pulses reach smaller FWHM relative to the T_2 limit, but their fitted contrast remains lower over the usable drive range. Their contrast-weighted operating windows in panel (c) therefore contract despite the high resolution. Pulse duration must consequently be selected jointly with contrast and experimental signal quality, rather than by linewidth alone.

Except for main-text Fig. 2, the numerical Hilbert space remains restricted to two levels. Those two-level calculations capture coherent drive dynamics, relaxation, and pure dephasing, but do not describe leakage, higher-level AC-Stark shifts, or multiphoton transitions. The high-drive resonance motion and leakage discussed in Sec. S7 are therefore evaluated with the explicit three-level model rather than inferred from these two-level calculations.

S7. THREE-LEVEL AC-STARK SHIFT OF THE f_{01} RESONANCE

We calculate the change of the central f_{01} resonance for a 40 μs constant square pulse and for 10 μs root-Lorentzian and echo-root-Lorentzian spectroscopy pulses. All calculations use the conventional drive–qubit detuning $\Delta = \omega_d - \omega_{01}$. The simulation retains $\{|g\rangle, |e\rangle, |f\rangle\}$ as dynamical levels, so population can occupy $|f\rangle$ during the pulse. In the weak-drive

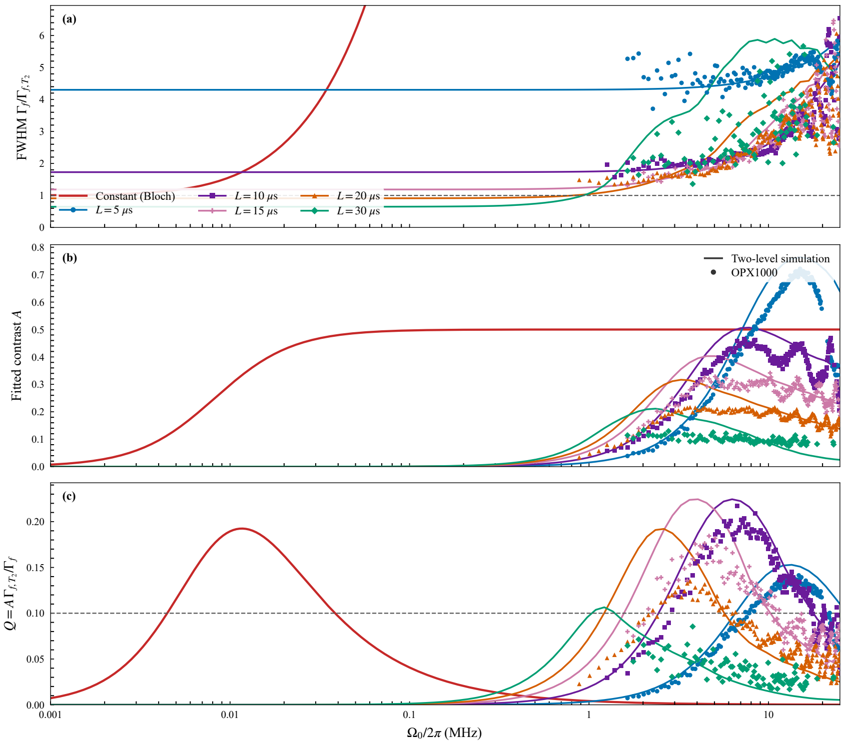

Figure S3. Resolution–contrast performance versus peak Rabi frequency up to 25 MHz for echo-root-Lorentzian pulses with $c = 0.005$ and $L = 5$, 10, 15, 20, and 30 μs. Colored continuous lines show two-level simulations, and colored markers show OPX1000 data. (a) Fitted FWHM in units of the coherence-limited reference, $\Gamma_f/\Gamma_{f,T_2}$, with the dashed line marking $\Gamma_f/\Gamma_{f,T_2} = 1$; (b) fitted contrast A; and (c) contrast-weighted resolution $Q = AR = A\Gamma_{f,T_2}/\Gamma_f$. The red curve is the constant-drive Bloch result, and the dashed line in (c) marks $Q = 0.10$.

interpretation, virtual $|e\rangle \leftrightarrow |f\rangle$ coupling produces the leading shift of f_{01}; this does not make $|f\rangle$ merely a virtual level in the numerical calculation. The plotted observable is the central excitation response $P_e + P_f$, not the position of the $g \leftrightarrow f$ transition. In this basis, the rotating-frame Hamiltonian of Eq. (S23) is

$$\frac{H(t)}{\hbar} = \begin{pmatrix} 0 & \Omega(t)/2 & 0 \\ \Omega(t)/2 & -\Delta & \Omega(t)/\sqrt{2} \\ 0 & \Omega(t)/\sqrt{2} & -2\Delta + \alpha \end{pmatrix}, \tag{S26}$$

where $\alpha = \omega_{12} - \omega_{01} < 0$. For a constant drive, virtual coupling between $|e\rangle$ and $|f\rangle$ gives the weak-drive result for a weakly anharmonic transmon [6, 8]:

$$\delta f_{01}^{(\mathrm{const})} \equiv \frac{\Delta_{\mathrm{res}}}{2\pi} \simeq -\frac{(\Omega_0/2\pi)^2}{2(\alpha/2\pi)}. \tag{S27}$$

Because $\alpha < 0$, the constant-pulse dressed f_{01} resonance moves to positive detuning. The exact values used here are obtained by diagonalizing Eq. (S26) at constant $\Omega(t) = \Omega_0$ and minimizing the dressed g–e gap.

For the shaped pulses, $\Omega(t) = \Omega_0 f(t)$ with the root-Lorentzian envelope of Eq. (S1), order $n = 1/2$, and endpoint amplitude $c = 0.002$. A useful weak-drive phase-average estimate is

$$\delta f_{01}^{(\mathrm{shape})} \simeq -\frac{(\Omega_0/2\pi)^2}{2(\alpha/2\pi)} \left\langle f^2 \right\rangle_t, \qquad \left\langle f^2 \right\rangle_t = \frac{2\sigma}{L} \tan^{-1}\left(\frac{L}{2\sigma}\right), \tag{S28}$$

where $\sigma = (L/2)/\sqrt{c^{-2} - 1} = 0.010000$ μs and $\langle f^2 \rangle_t = 0.003138$. The echo phase inversion changes $f(t) \to -f(t)$ in the second half and therefore does not reverse this quadratic Stark term. Consequently, the root-Lorentzian and echo-root-Lorentzian pulses have the same leading phase-average estimate.

The experimentally visible feature of a finite pulse need not coincide with Eq. (S28), because the final population also contains coherent finite-pulse interference. We therefore distinguish both quantities. For the ordinary root-Lorentzian pulse, the operational center is the local maximum of $P_e + P_f$ within $|\Delta|/(2\pi) \leq 0.5$ MHz. For the echo-root-Lorentzian pulse it is the corresponding central minimum. The spectra are evaluated on a 0.05 MHz grid and the extrema are refined with cubic interpolation. The central-feature extraction is evaluated for $\Omega_0/(2\pi) \geq 20$ MHz, where the feature has sufficient contrast.

The small negative position of the echo-root symmetry center at the largest drive is an operational finite-pulse feature position, not a negative instantaneous AC-Stark shift. It results from the interference defining the echo spectrum. The phase-average Stark term remains positive for both shaped protocols.

The time-domain maps solve the three-level Lindblad equation with $T_1 = 51.24$ μs, $T_\phi = 7.87$ μs, an initial $|g\rangle$ state, 31 peak Rabi frequencies from 0 to 60 MHz, and 101 detuning points over the full 100 MHz domain, from -50 to 50 MHz. Additional 401-point scans resolve the central 20 MHz span and the 1 MHz Lorentzian zooms. The model uses the rotating-wave approximation, so it does not include a counter-rotating Bloch–Siegert shift [9].

S7.1. AC-Stark scale over the common drive span

The AC-Stark calculation above predicts a clear separation between the instantaneous shift produced by a constant drive and that produced by the Lorentzian-derived envelopes. At the common upper drive $\Omega_0/(2\pi) = 50$ MHz, the weak-drive constant-pulse estimate from Eq. (S27) is approximately 5.8 MHz. For the 20 μs root-Lorentzian envelope with $c = 0.001$, multiplication by $\langle f^2 \rangle_t = 0.001570$ reduces the phase-average estimate to 9.1 kHz. Fig. 5 of the Letter shows this separation over the full calculated drive range. The echo phase inversion does not cancel the quadratic Stark term; rather, the small time-averaged squared envelope suppresses it, while finite-pulse interference determines the experimentally visible central minimum.

S7.2. Accumulated-phase compensation of drive-induced center motion

The phase inversion at the pulse midpoint suppresses the resonant pulse area, but it does not cancel the AC-Stark shift. The reason is that the leading shift is proportional to the drive intensity, $\Omega_I^2(t)$, which is unchanged by $\Omega_I \to -\Omega_I$. To compensate this even-in-amplitude contribution, we add an opposing quadratic correction to the instantaneous drive detuning,

$$\frac{\Delta_{\mathrm{corr}}(t)}{2\pi} = \kappa \left[\frac{\Omega_I(t)}{2\pi} \right]^2, \tag{S29}$$

where κ has units of MHz^{-1}. With all cyclic frequencies expressed in MHz, weak-drive perturbation theory gives the natural starting value

$$\kappa_{\mathrm{theory}} \simeq -\frac{1}{2(\alpha/2\pi)} = +0.00231\ \mathrm{MHz}^{-1} \qquad \left[\frac{\alpha}{2\pi} = -216.0\ \mathrm{MHz} \right]. \tag{S30}$$

The coefficient is then refined numerically because finite pulse duration, strong driving, decoherence, and three-level dynamics are not fully described by the weak-drive estimate. For a fixed duration, a single value of κ is used for the whole amplitude–detuning map. It is not recomputed from the fitted f_{10} at each amplitude and does not depend on the trial spectroscopy detuning. The optimum can depend on pulse duration and other waveform details, however; the duration comparison below therefore optimizes κ separately for each duration rather than asserting one universal coefficient. Once calibrated for a specified device and waveform, the coefficient can be applied while the carrier frequency is being searched.

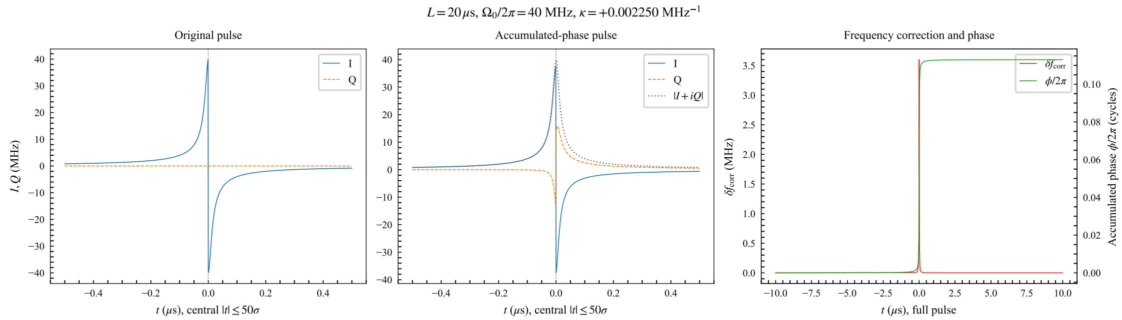

Figure S4. Hardware-realizable accumulated-phase correction for a 20 μs echo-root-Lorentzian pulse at $\Omega_0/(2\pi) = 40$ MHz. From left to right: the original real-valued echo pulse, the I and Q components after multiplication by $\exp[-i\phi(t)]$, and the quadratic frequency addition together with its accumulated phase. The pulse magnitude is unchanged.

On hardware, a separate sample-by-sample detuning control is not required. We integrate Eq. (S29) into the accumulated phase

$$\phi(t) = \int_{t_0}^{t} \Delta_{\rm corr}(t')\, dt' = 2\pi\kappa \int_{t_0}^{t} \left[\frac{\Omega_I(t')}{2\pi}\right]^2 dt', \tag{S31}$$

and rotate the programmed complex envelope according to

$$s_{\rm corr}(t) = s_0(t) \exp[-i\phi(t)]. \tag{S32}$$

With the complex-envelope and detuning conventions used here, $d\phi/dt = \Delta_{\rm corr}(t)$ produces precisely the additional instantaneous detuning in Eq. (S29). Using $\exp[-i\Delta_{\rm corr}(t)t]$ instead would not be equivalent: its instantaneous frequency contains the unwanted term $t\, d\Delta_{\rm corr}/dt$. The phase must therefore be the time integral of the desired frequency correction.

This construction has the same physical intuition as the phase correction in DRAG [10, 11]. In DRAG, a derivative quadrature suppresses leakage and a drive-dependent phase or frequency update compensates the accompanying dynamical phase. Here we deliberately set the DRAG coefficient to $\beta = 0$: no derivative quadrature is added, and the only change to the echo-root-Lorentzian waveform is the accumulated phase proportional to $\int \Omega_I^2 dt$. Because the midpoint echo changes the sign of Ω_I but not Ω_I^2, the phase accumulator remains continuous through the $0/\pi$ inversion. Fig. S4 shows the original and corrected envelopes and the relation between the localized frequency correction and its accumulated phase.

The reproducible three-level calculation is generated by a source-controlled workflow for accumulated-phase duration reports. Numerical arrays are retained alongside the generated analysis data. The calculation uses a 20 μs pulse with cutoff $c = 0.001$, echo inversion enabled, $\beta = 0$, $\alpha/(2\pi) = -216.0$ MHz, $T_1 = 51.24$ μs, $T_{2,\text{eff}} = 7.31$ μs; the corresponding pure-dephasing time is $T_\phi = 7.87$ μs. The final populations P_g, P_e, and P_f are calculated over peak Rabi frequencies from 0 to 80 MHz and detunings from -1 to 1 MHz.

For each candidate κ, the notebook calculates a compact map over $\Omega_0/(2\pi) = 20$–80 MHz. The central minimum of P_e is identified within $|\Delta|/(2\pi) \leq 0.35$ MHz and refined with a local quadratic fit. The optimization target is its RMS displacement from the bare transition,

$$\mathcal{L} = \sqrt{\langle \delta_0^2(\Omega_0) \rangle_{\Omega_0}}, \tag{S33}$$

while the maximum P_f is retained as an independent leakage diagnostic. A coarse scan over $\kappa = -0.005$ to 0.005 MHz^{-1} is followed by a fine scan around the best coarse value. As shown in Fig. S5(a), the optimum for this model is

$$\kappa = +0.00225 \text{ MHz}^{-1}. \tag{S34}$$

The displayed digits identify the selected point of the numerical scan; they do not imply corresponding experimental or model accuracy. This value is of the same order as the weak-drive estimate in Eq. (S30). Within this three-level rotating-wave model, the fitted operational central minimum then has an RMS displacement of 0.14 kHz from the bare transition on the optimization grid and a maximum absolute displacement of 0.41 kHz over the dense simulated map, compared with a 13.52 kHz maximum offset without correction. These are numerical residuals of the optimized operational spectral center, rather than a claim of sub-kilohertz physical accuracy or a direct determination of the instantaneous AC-Stark shift. Quantitative validation would require time-step and Hilbert-space convergence, inclusion or bounding of counter-rotating (Bloch–Siegert) contributions, and comparison with device-specific calibration and control-bandwidth limitations. Thus, κ is not yet an experimentally calibrated correction parameter.

The duration comparison is shown in Fig. S5(b). In all three columns, the accumulated-phase correction straightens the central P_e feature without changing the coherence parameters or amplitude and detuning grids. For the 20 μs calculation, the maximum P_f over the full map decreases from 0.0974 to 0.0861. The FWHM remains finite. In units of $\Gamma_{f,T_{2,\text{eff}}} = 1/(\pi T_{2,\text{eff}})$, its simulated range changes from 1.09–5.87 without correction to 1.09–5.32 with correction. The cost in Eq. (S33) constrains the resonance position, not population ripple, so residual coherent oscillations remain.

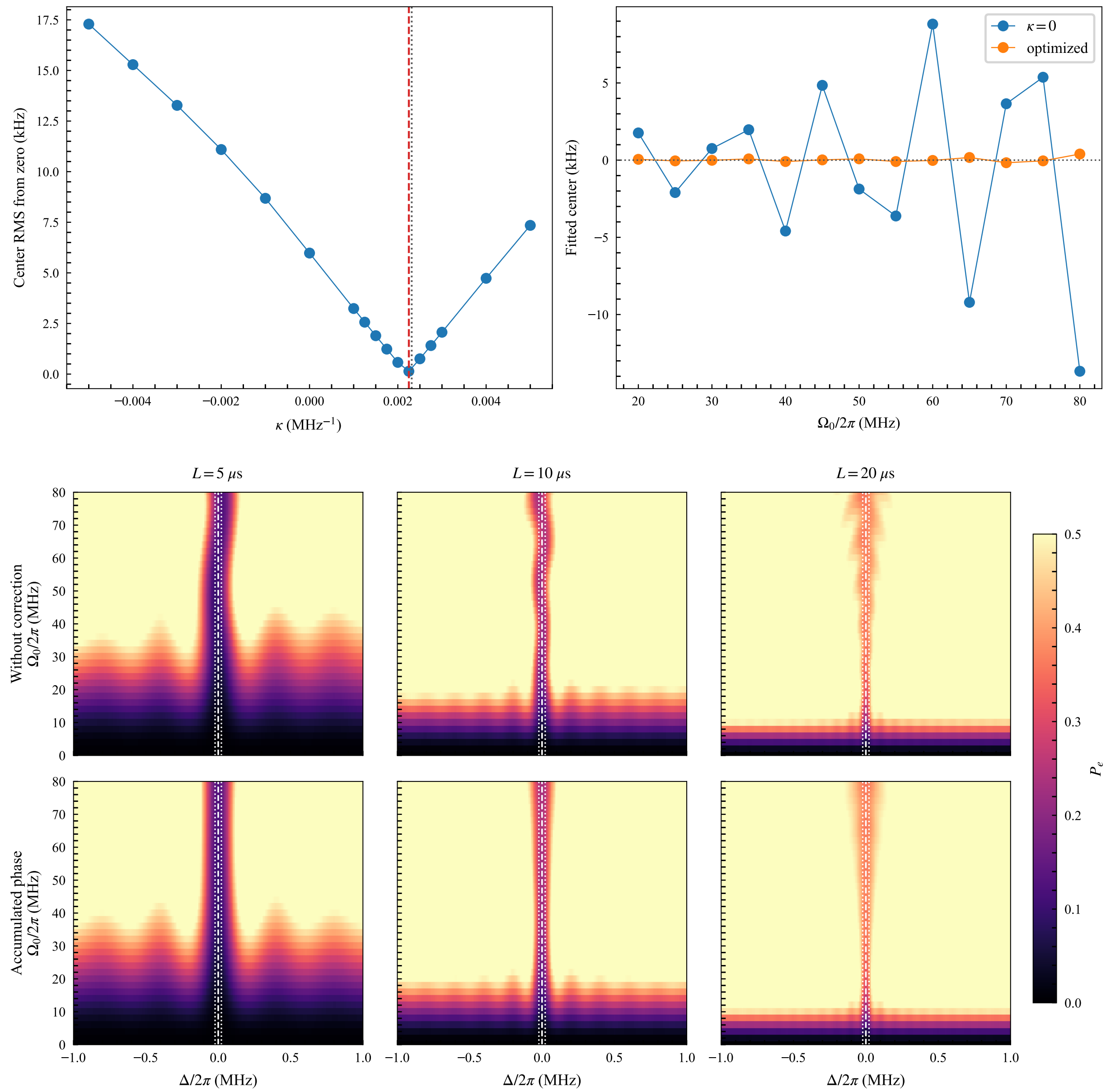

Figure S5. Three-level optimization of the echo-root-Lorentzian response. (a) Quadratic Stark-correction scan. The left panel shows the RMS fitted-center displacement from the bare transition versus κ; the dashed line marks the selected coefficient. The right panel compares the fitted central minimum without the correction and at the optimized value. (b) Excited-state population P_e over the full -1 to 1 MHz detuning range for pulse lengths $L = 5$, 10, and 20 μs. The upper row is uncorrected and the lower row uses the optimized accumulated phase for each duration, always with $\beta = 0$. The common color scale is clipped at $P_e = 0.5$ for visibility. Dashed lines mark the bare transition and paired dotted lines mark the $T_{2,\text{eff}}$-limited FWHM bounds.

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